FOURIER–MUKAI TRANSFORMS COMMUTING WITH FROBENIUS

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Abstract. We show that a Fourier–Mukai equivalence between smooth projective varieties of characteristic p which commutes with either pushforward or pullback along Frobenius is a composition of shifts, isomorphisms, and tensor product with invertible sheaves whose $(p-1)$th tensor power is trivial.

1. Introduction

Let X be a smooth projective variety over an algebraically closed field k of positive characteristic p. We write $\mathrm{D}^b(X)$ for the bounded derived category of coherent sheaves on X, viewed as a k–linear triangulated category. The *absolute Frobenius morphism* of X is the morphism $\mathrm{F}_X : X \to X$ induced by the pth power map $\mathcal{O}_X \to \mathcal{O}_X$, given on local sections by $f \mapsto f^p$. The morphism F_X is finite and flat, so both pushforward and pullback along F_X are exact. We consider the endofunctors

$$\mathrm{F}_{X*}, \mathrm{F}_X^* \colon \mathrm{D}^b(X) \to \mathrm{D}^b(X).$$

Here, as in the rest of this note, we denote derived functors by the same symbols as their underived counterparts. We remark that the functors F_{X*} and F_X^* are not k–linear; instead, they are *k–semilinear* with respect to the Frobenius morphism of k. Let Y be another smooth projective variety over k, and let

$$\Phi \colon \mathrm{D}^b(X) \to \mathrm{D}^b(Y)$$

be a k–linear Fourier–Mukai equivalence. We say that Φ *commutes with* F_* (resp. Φ *commutes with* F^*) if the diagram

$$\begin{array}{ccc} \mathrm{D}^b(X) & \xrightarrow{\Phi} & \mathrm{D}^b(Y) \\ {\scriptstyle \mathrm{F}_{X*}}\downarrow & & \downarrow{\scriptstyle \mathrm{F}_{Y*}} \\ \mathrm{D}^b(X) & \xrightarrow{\Phi} & \mathrm{D}^b(Y) \end{array} \qquad \text{resp.} \qquad \begin{array}{ccc} \mathrm{D}^b(X) & \xrightarrow{\Phi} & \mathrm{D}^b(Y) \\ {\scriptstyle \mathrm{F}_X^*}\downarrow & & \downarrow{\scriptstyle \mathrm{F}_Y^*} \\ \mathrm{D}^b(X) & \xrightarrow{\Phi} & \mathrm{D}^b(Y) \end{array}$$

commutes up to a natural isomorphism. Here are some examples of equivalences which commute with both F_* and F^*:

(1) the shift functor $[n] \colon \mathrm{D}^b(X) \to \mathrm{D}^b(X)$ for any $n \in \mathbf{Z}$,
(2) $f_* \colon \mathrm{D}^b(X) \to \mathrm{D}^b(Y)$, where $f \colon X \xrightarrow{\sim} Y$ is an isomorphism over k, and

(3) $L \otimes \_\colon \mathrm{D}^b(X) \to \mathrm{D}^b(X)$, where L is an invertible sheaf on X such that $L^{\otimes p-1} \cong \mathcal{O}_X$.

Indeed, the shift functor commutes with any map of triangulated categories (by definition). The absolute Frobenius has the property that $\mathrm{F}_Y \circ f = f \circ \mathrm{F}_X$ for any map of schemes $f : X \to Y$. Finally, if L is an invertible sheaf on X, then $\mathrm{F}_X^* L \cong L^{\otimes p}$. Therefore if $L^{\otimes p-1} \cong \mathcal{O}_X$, then we have $\mathrm{F}_X^* L \cong L$, and so for any $E \in \mathrm{D}^b(X)$ we obtain isomorphisms

$$\mathrm{F}_X^*(L \otimes E) \cong \mathrm{F}_X^* L \otimes \mathrm{F}_X^* E \cong L \otimes \mathrm{F}_X^* E$$

and

$$\mathrm{F}_{X*}(L \otimes E) \cong \mathrm{F}_{X*}(\mathrm{F}_X^* L \otimes E) \cong L \otimes \mathrm{F}_{X*} E$$

which are functorial in E.

In this note we will show that these are in fact the only examples.

Theorem 1.1. *If $\Phi\colon \mathrm{D}^b(X) \to \mathrm{D}^b(Y)$ is a Fourier–Mukai equivalence which commutes with F_* or with F^*, then Φ is a composition of functors of the above form. More precisely, there exists an isomorphism $f\colon X \xrightarrow{\sim} Y$ over k, an invertible sheaf L on X such that $L^{\otimes p-1} \cong \mathcal{O}_X$, and an integer n such that*

$$\Phi(\_) \cong f_*(L \otimes \_)[n].$$

The key step is Proposition 3.3, which characterizes the shifts of structure sheaves of closed points of X among all objects of $\mathrm{D}^b(X)$ in terms of the k–linear triangulated category structure on $\mathrm{D}^b(X)$ together with the Frobenius endofunctor $\mathrm{F}_{X*}\colon \mathrm{D}^b(X) \to \mathrm{D}^b(X)$.

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2. Equivalences preserving supports

Let X be a smooth variety over an algebraically closed field k (of arbitrary characteristic). The *support* of a coherent sheaf E on X is the closed subscheme of X cut out by the ideal sheaf $I \subset \mathcal{O}_X$ which is the kernel of the action map

$$\mathcal{O}_X \to \mathcal{E}nd_{\mathcal{O}_X}(E).$$

Equivalently, the support of E is the minimal closed subscheme $Z \subset X$ such that E is the pushforward of a coherent sheaf on Z. The *support* of a complex $E \in \mathrm{D}^b(X)$ is the minimal closed subscheme of X which contains the support of every cohomology sheaf of E. If E is a coherent sheaf or an object of $\mathrm{D}^b(X)$, we define the *reduced support* of E to be the reduced subscheme of X underlying the support.

We make the following definition.

Definition 2.1. An object $E \in \mathrm{D}^b(X)$ is *point–like* if

(1) $\mathrm{Hom}_{\mathrm{D}^b(X)}(E, E[i]) = 0$ for $i < 0$ and

(2) $\mathrm{Hom}_{\mathrm{D}^b(X)}(E, E) \cong k$.

If $x \in X$ is a closed point, then any shift $k(x)[n]$ is a point–like object of $\mathrm{D}^b(X)$. In general, there may be point–like objects with positive dimensional support. The following result shows however that every point–like object with 0–dimensional support is of this form.

Lemma 2.2. *If $E \in \mathrm{D}^b(X)$ is a point–like object with 0–dimensional support, then $E \cong k(x)[n]$ for some closed point $x \in X$ and integer n.*

Proof. See [1, Lemma 4.5]. □

Suppose now that X is smooth and projective. Let Y be another smooth projective variety over k and let $\Phi : \mathrm{D}^b(X) \to \mathrm{D}^b(Y)$ be a Fourier–Mukai equivalence with kernel $P \in \mathrm{D}^b(X \times Y)$.

Proposition 2.3. *Suppose that, for every closed point $x \in X$, the support of the complex $\Phi(k(x)) \in D^b(Y)$ has dimension 0. Then the support of P is the graph of an isomorphism $f : X \xrightarrow{\sim} Y$, and there exists an integer n and an invertible sheaf L on X such that*

$$\Phi(\_) \cong f_*(L \otimes \_)[n].$$

Proof. Consider a closed point $x \in X$. Then $k(x)$ is point-like, so $\Phi(k(x))$ is as well. By assumption, $\Phi(k(x))$ also has zero dimensional support, so by Lemma 2.2 we have $\Phi(k(x)) \cong k(y)[n]$ for some closed point $y \in Y$ and integer n. By semicontinuity the integer n must be independent of the point x [1, Corollary 6.14]. We conclude by applying [1, Corollary 5.23] to the functor $[-n] \circ \Phi$. □

3. Characterizing points using Frobenius

Let X be a smooth variety over an algebraically closed field k of characteristic $p > 0$. Let $Z \subset X$ be a reduced and irreducible closed subscheme with generic point η_Z. We say that a coherent sheaf E on X is *properly supported on Z* if either Z is an irreducible component of the reduced support of E or E vanishes generically on Z. Equivalently, E is properly supported on Z if the restriction $E \otimes_{\mathcal{O}_X} \mathcal{O}_{X,\eta_Z}$ of E to the local scheme $\operatorname{Spec} \mathcal{O}_{X,\eta_Z}$ has finite length. If E is a coherent sheaf on X which is properly supported on Z, we define

$$\mathrm{rk}_Z(E) := \mathrm{len}_{\mathcal{O}_{X,\eta_Z}}(E \otimes_{\mathcal{O}_X} \mathcal{O}_{X,\eta_Z}).$$

The key property of this quantity that we will use is that it is additive in short exact sequences of coherent sheaves properly supported on Z.

Lemma 3.1. *If $Z \subset X$ is a reduced and irreducible closed subvariety of dimension d and E is a coherent sheaf on X which is properly supported on Z, then we have*

$$\mathrm{rk}_Z(\mathrm{F}_{X*}E) = p^d \, \mathrm{rk}_Z(E).$$

Proof. Let E be a coherent sheaf properly supported on Z. Let Z' be the support of E. We first consider the special case when $Z' = Z = X$. By passing to an open subscheme, we may assume that E is free, and hence further reduce to the case when $E = \mathcal{O}_X$. The result then follows from the fact that the absolute Frobenius of a smooth k-variety of dimension n has degree p^n.

Next we consider the special case when $Z' \subset Z$. By passing to an open subscheme we may assume that $Z' = Z$ and that Z is smooth. Let $i\colon Z \hookrightarrow X$ be the inclusion. We have a commutative diagram

$$\begin{tikzcd} Z \arrow[r, "\mathrm{F}_Z"] \arrow[d, hook, "i"'] & Z \arrow[d, hook, "i"] \\ X \arrow[r, "\mathrm{F}_X"] & X. \end{tikzcd}$$

This implies that

$$\mathrm{F}_{X*}E = \mathrm{F}_{X*}(i_*i^*E) = i_*(\mathrm{F}_{Z*}(i^*E))$$

and hence

$$i^*(\mathrm{F}_{X*}E) = i^*i_*(\mathrm{F}_{Z*}(i^*E)) = \mathrm{F}_{Z*}(i^*E).$$

The previous case gives the equality $\mathrm{rk}_Z(\mathrm{F}_{Z*}(i^*E)) = p^d \,\mathrm{rk}_Z(i^*E)$, and so we have

$$\mathrm{rk}_Z(\mathrm{F}_{X*}E) = \mathrm{rk}_Z(i^*\mathrm{F}_{X*}E) = \mathrm{rk}_Z(\mathrm{F}_{Z*}(i^*E)) = p^d \,\mathrm{rk}_Z(i^*E) = p^d \,\mathrm{rk}_Z(E).$$

Finally, we consider the general case. Let I be the ideal sheaf of Z. If E is properly supported on Z, then for any integer $m \geq 1$ the coherent sheaf I^mE is also properly supported on Z. Furthermore, for all sufficiently large integers m, we have that $\mathrm{rk}_Z(I^mE) = 0$ (equivalently, I^mE vanishes generically on Z). We induct on m the claimed statement for those coherent sheaves E which are properly supported on Z and satisfy $\mathrm{rk}_Z(I^mE) = 0$. For the base case, we observe that if E is properly supported on Z and satisfies $\mathrm{rk}_Z(IE) = 0$, then the support of E is contained in Z, and we are in the situation already dealt with above. For the induction step, suppose that $\mathrm{rk}_Z(I^{m+1}E) = 0$, and consider the short exact sequence

$$0 \to IE \to E \to E/IE \to 0.$$

The left and right terms are again properly supported on Z, and moreover each of their products with I^m has $\mathrm{rk}_Z = 0$. Applying F_{X*} and using the additivity of rk_Z and the induction hypothesis, we get

$$\begin{aligned} \mathrm{rk}_Z\left(\mathrm{F}_{X*}E\right) &= \mathrm{rk}_Z\left(\mathrm{F}_{X*}IE\right) + \mathrm{rk}_Z\left(\mathrm{F}_{X*}(E/IE)\right) \\ &= p^d \,\mathrm{rk}_Z(IE) + p^d \,\mathrm{rk}_Z\left(E/IE\right)) \\ &= p^d \,\mathrm{rk}_Z(E). \end{aligned}$$

□

Lemma 3.2. *If $E \in \mathrm{D}^b(X)$ is a nonzero object such that $E \cong \mathrm{F}_{X*}E$, then E has 0-dimensional support.*

Proof. If $E \cong \mathrm{F}_{X*}E$, then also $\mathscr{H}^i(E) \cong \mathscr{H}^i(\mathrm{F}_{X*}E) = \mathrm{F}_{X*}\mathscr{H}^i(E)$ for each integer i. We therefore reduce to the case when E is a nonzero coherent sheaf. Let Z be the reduction of an irreducible component of the support of E. Then E is properly supported on Z and $\mathrm{rk}_Z(E) \neq 0$. By Lemma 3.1, we have

$$\mathrm{rk}_Z(E) = \mathrm{rk}_Z(\mathrm{F}_{X*}E) = p^d \,\mathrm{rk}_Z(E)$$

where d is the dimension of Z. It follows that $d = 0$. □

Combining the above results, we obtain the following characterization of the shifts of structure sheaves of points in $\mathrm{D}^b(X)$.

Proposition 3.3. *Let $E \in \mathrm{D}^b(X)$ be an object. The following are equivalent.*
(1) $E \cong k(x)[n]$ for some closed point $x \in X$ and integer n.
(2) E is point–like and $E \cong \mathrm{F}_{X*}E$.

Proof. (1) implies (2) is immediate. We prove that (2) implies (1). By Lemma 3.2, if $E \cong \mathrm{F}_{X*}E$ then E has 0-dimensional support. The result then follows by applying Lemma 2.2. □

4. Proof of Theorem 1.1

We recall the notation: X and Y are smooth projective varieties over an algebraically closed field k of characteristic $p > 0$ and $\Phi\colon \mathrm{D}^b(X) \to \mathrm{D}^b(Y)$ is a Fourier–Mukai equivalence.

Lemma 4.1. *The equivalence Φ commutes with F^* if and only if it commutes with F_*.*

Proof. We have the adjunction

$$\mathrm{Hom}_{\mathrm{D}^b(X)}(\mathrm{F}_X^*E, G) \cong \mathrm{Hom}_{\mathrm{D}^b(X)}(E, \mathrm{F}_{X*}G)$$

for objects $E, G \in \mathrm{D}^b(X)$. Because Φ is an equivalence, this gives rise to isomorphisms

$$\mathrm{Hom}_{\mathrm{D}^b(Y)}(\Phi(\mathrm{F}_X^*E), \Phi(G)) \cong \mathrm{Hom}_{\mathrm{D}^b(Y)}(\Phi(E), \Phi(\mathrm{F}_{X*}G))$$

which are functorial in E and G. Suppose that $\mathrm{F}_Y^* \circ \Phi \cong \Phi \circ \mathrm{F}_X^*$. Then we obtain functorial isomorphisms

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{D}^b(Y)}(\Phi(E), \mathrm{F}_{Y*}\Phi(G)) &\cong \mathrm{Hom}_{\mathrm{D}^b(Y)}(\mathrm{F}_Y^*\Phi(E), \Phi(G)) \\
&\cong \mathrm{Hom}_{\mathrm{D}^b(Y)}(\Phi(\mathrm{F}_X^*E), \Phi(G)) \\
&\cong \mathrm{Hom}_{\mathrm{D}^b(Y)}(\Phi(E), \Phi(\mathrm{F}_{X*}G)).
\end{aligned}$$

As Φ is an equivalence, Yoneda's Lemma implies that $\mathrm{F}_{Y*} \circ \Phi \cong \Phi \circ \mathrm{F}_{X*}$. The reverse implication is similar. □

Proof of Theorem 1.1. By Lemma 4.1, it suffices to consider the case when Φ commutes with F_*. For a closed point $x \in X$, we have that $\Phi(k(x))$ is point–like, and furthermore

$$\Phi(k(x)) \cong \Phi(\mathrm{F}_{X*}k(x)) \cong \mathrm{F}_{X*}\Phi(k(x)).$$

Proposition 3.3 implies that $\Phi(k(x)) \cong k(y)[n]$ for some closed point $y \in Y$ and some integer n, and in particular $\Phi(k(x))$ has zero dimensional support. By Proposition 2.3, there exists an isomorphism $f\colon X \xrightarrow{\sim} Y$, an integer n, and an invertible sheaf L on X such that

$$\Phi(\_) \cong f_*(L \otimes \_)[n].$$

It remains to show that $L^{\otimes p-1} \cong \mathcal{O}_X$. To see this, we note that shifts and pushforwards along isomorphisms always commute with both F^* and F_*. It follows that tensoring with L commutes with F_*, hence also with F^*, and therefore we have

$$L \otimes \mathrm{F}_X^* E \cong \mathrm{F}_X^*(L \otimes E)$$

for every $E \in \mathrm{D}^b(X)$. In particular, taking $E = \mathcal{O}_X$ we conclude that $L^{\otimes p-1} \cong \mathcal{O}_X$. □

References

[1] D. Huybrechts. *Fourier-Mukai transforms in algebraic geometry*. Oxford Mathematical Monographs. The Clarendon Press, Oxford University Press, Oxford, 2006.

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